

Why did direct cost per order rise to ₹115?

Eternal Ltd disclosed a **2.0% QoQ rise in Blinkit's direct cost per order to ₹115** in Q1 FY27, costing **10bps of contribution margin**. This case diagnoses the operational failure mode behind it, using 588,832 Play Store reviews matched to the same disclosed fiscal quarters.

- Q4 FY26 → Q1 FY27
- UNSUPERVISED DISCOVERY
- TWO-PROPORTION Z-TEST
- BONFERRONI + BH
- CROSS-RUN REPLICATION

331M	2,443	₹17,132cr	₹518
ORDERS Q1 FY27	DARK STORES +200 NET NEW	NET ORDER VALUE, Q1 FY27	AVERAGE ORDER VALUE (₹525 QOQ)

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THE ANSWER, UP FRONT

Cold-chain failure, **not** service, speed or cancellations

Melted & spoiled-item complaints were the only failure mode to more than double QoQ, and the only one to survive every robustness test across two independent pipeline runs.



Share of negative reviews mentioning melted / spoiled cold-chain items · RR 2.37× · $p_{adj} = 2.7 \times 10^{-19}$ · Cohen's h = 0.103

WHY IT SHOWS UP IN DIRECT COST PER ORDER



ALL CATEGORIES THAT SHIFTED SIGNIFICANTLY (OF 25)

▲ Cold-chain / melted items	2.37×
▲ Produce quality (rotten)	1.20×
▲ General dissatisfaction	1.15×
▼ Delivery / handling charges	0.84×

DISCLOSED TRIGGER

₹115 / order
+2.0% QoQ · 10bps margin decline

SCALE OF IMPACT

₹74.6 crore
Incremental direct cost, Q1 FY27

EVIDENCE BASE

588,832 reviews
25 complaint categories discovered

VERDICT

1 of 25 categories
Significant, robust & replicated

Cold-chain failure is the only validated driver of the ₹115 cost rise. Customer service, delivery speed and cancellations are not.

588,832 reviews were scraped and 25 complaint categories were discovered in the negative band. Four shifted significantly QoQ. Only **melted / spoiled cold-chain items** more than doubled, survived version-mix reweighting, and replicated in a second independent run. Most of that rise is a summer pattern that repeats annually, which makes it predictable rather than mysterious.

₹115/order

DIRECT COST • Q1 FY27

+2.0% QoQ from ₹112.75

10_{bps}

CONTRIBUTION MARGIN DECLINE

5.4% → 5.3% of NOV

₹74.6_{cr}

INCREMENTAL DIRECT COST

₹2.25 × 331M orders

2.37_×

COLD-CHAIN COMPLAINT RISE

0.90% → 2.13% of negatives

What the data says

Q4 FY26 → Q1 FY27 • NEGATIVE BAND

Cold-chain / melted items ice cream, melted, chilled, cold		0.90% 2.13%	2.37 _×	PRIORITY
Produce quality (rotten) vegetables, rotten, fruits, fresh		3.03% 3.62%	1.20 _×	UNSTABLE
General dissatisfaction blinkit, experience, order, money		9.69% 11.17%	1.15 _×	REPLICATES
Delivery / handling charges charges, handling, extra, high		8.01% 6.72%	0.84 _×	NOT ROBUST

21 categories showed no significant QoQ shift, including customer care, late delivery, rider behaviour, cancellations, refunds and wrong/missing items. All cleared the 30-review sample floor, so these are genuine nulls, not underpowered tests.

What to do about it

RANKED BY IMPACT / EFFORT

1

Insulated packaging for frozen & dairy SKUs at pack-out
Interrupts melt-in-transit, the mechanism named in every cold-chain complaint sampled. Lowest effort, highest impact.

2

Dark-store freezer uptime & capacity audit
The 10–15 min model relies on dark-store cold storage, not refrigerated transit. A freezer running warm fails upstream of the rider.

3

Separate dwell-time SLA for frozen items in mixed baskets
Stops frozen picks queuing behind ambient items during multi-item order assembly.

SIZING THE OPPORTUNITY • SHARE OF THE ₹74.6CR RISE

If cold-chain explains 10%

₹7.5 cr

Conservative anchor • 25%

₹18.7 cr

If cold-chain explains 50%

₹37.3 cr

WHY THIS IS CREDIBLE

The finding survived **two independent pipeline runs** in separate environments, and two of the four candidate findings did not.

WHY IT WAS MISSED

Cold-chain isn't a standard cost bucket. It only surfaced because categories were **discovered, not assumed**.

HOW MUCH IS JUST SUMMER

Extending the series back three years shows this spike recurs every Apr–Jun. About **80% of the rise is the normal seasonal pattern**, and the remaining 20% is deterioration on top of it.

Sources: Eternal Ltd Q1 FY27 disclosure (direct cost/order, contribution margin). Q1 FY27 investor deck via Business Standard & Lapaas Voice (331M orders, +21% QoQ). **Review analysis:** own pipeline over 588,832 scraped Play Store reviews (209,680 in the negative band), BERTopic discovery, two-proportion z-test with Bonferroni/BH correction, version-mix robustness reweighting. n = 15,885 (Q4 FY26) vs 19,440 (Q1 FY27) negative reviews.

Direct cost rose ₹718cr in one quarter. Only ₹74.6cr of it is an efficiency loss. That is the part worth diagnosing.

Blinkit grew orders 20.8% QoQ, so most of the cost increase is simply volume. Stripping that out isolates the **rate effect**: the ₹2.25 per order that Eternal disclosed as a 2.0% QoQ rise, and the 10bps of contribution margin it consumed.

Decomposing the ₹718cr increase

VOLUME VS RATE EFFECT

TOTAL DIRECT-COST INCREASE, Q4 FY26 → Q1 FY27

₹643.8cr

89.6% · VOLUME EFFECT

57.1M more orders × ₹112.75 old rate

₹718.4cr

₹74.6cr

10.4% · RATE

331M orders × ₹2.25 rate rise

Growth explains nearly nine-tenths of the increase and is a deliberate strategic cost. The remaining ₹74.6cr is unit-economics deterioration, cost that would not have appeared had per-order efficiency simply held flat.

THE QUESTION THIS CASE ANSWERS

Which operational failure mode drove the ₹2.25 per-order rate rise, and is it large enough to be worth fixing?

₹112.75

Q4 FY26
IMPLIED

+

₹2.25

RATE RISE
+2.0% QoQ

=

₹115.00

Q1 FY27
DISCLOSED

×

331M

ORDERS
Q1 FY27

What Eternal disclosed

BLINKIT SEGMENT

METRIC	Q4 FY26	Q1 FY27	Δ QOQ
Direct cost / order the disclosed trigger	₹112.75 implied	₹115.00	+2.0%
Contribution margin level not consistently reported	n/a	n/a	−10bps
Net order value	₹14,386cr	₹17,132cr	+19.1%
Orders	273.9M	331.0M	+20.8%
Average order value	₹525	₹518	−1.3%
Take rate	26.9%	27.5%	+60bps
Adjusted EBITDA	₹37cr	₹102cr	+176%
Total direct cost orders × cost/order	₹3,088cr	₹3,806cr	+23.3%

Q4 FY26 direct cost per order is not separately disclosed, so ₹112.75 is implied by reversing the reported +2.0% QoQ. Net order value cross-checks against orders × AOV to within 0.1% in both quarters. Contribution-margin **level** is reported inconsistently across secondary sources, so only the 10bps delta is used.

WHY 10BPS MATTERS

Take rate **improved 60bps** while direct cost per order **worsened 2.0%**. Blinkit monetised better and executed worse in the same quarter. With the segment only just adjusted-EBITDA positive, the slip is an operations problem, not a pricing one.

WHAT "DIRECT COST PER ORDER" CAN CONTAIN

Cost lines a complaint category could plausibly map to:

Last-mile delivery

Dark-store operations

Inventory wastage

Packaging

Replacements & refunds

Customer support

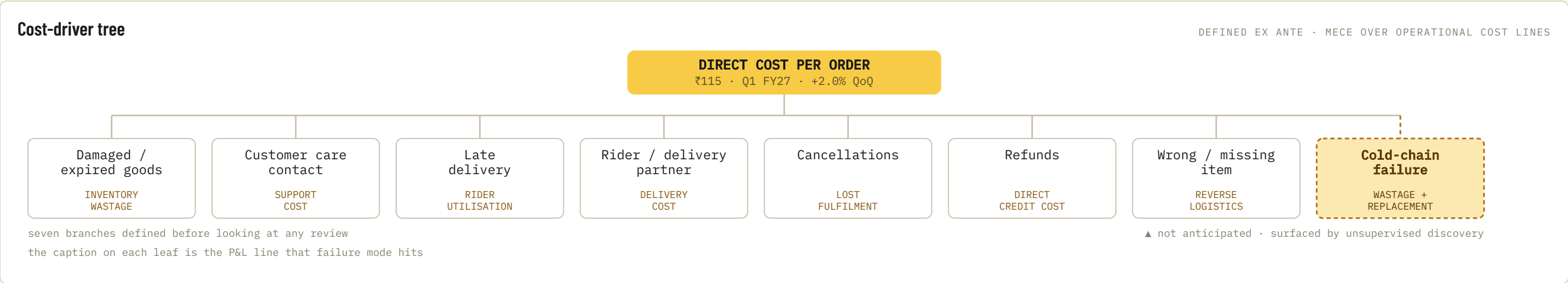
WHY THIS IS HARD TO DIAGNOSE EXTERNALLY

Eternal discloses direct cost per order as a **single blended number**, with no wastage, delivery or support split. An independent signal is needed to attribute the movement.

Sources: Eternal Ltd Q1 FY27 results: direct cost/order ₹115, +2.0% QoQ, 10bps contribution-margin decline. Order volume, AOV and dark-store counts from Q1 FY27 investor deck coverage (Business Standard, Lapaas Voice, INDmoney), Jul–Aug 2026. **Derived by author:** implied Q4 FY26 cost/order, implied GOV, total direct cost, and the volume/rate decomposition.

Direct cost per order breaks into seven operational failure modes. The one that mattered was in none of them.

The seven branches below cover every cost line an operational failure can hit, without overlapping each other. All seven were written down **before** any review data was examined. Cold-chain failure was not anticipated. It surfaced only because complaint categories were discovered from the corpus rather than assumed.



Reviews cannot measure cost. They can measure **which failure mode is getting worse**, independently of the teams being diagnosed.

588,832 Blinkit Play Store reviews were scraped and banded by rating. The analysis uses the **composition of the negative band**: what share of complaints are about a given failure, rather than absolute complaint volume, which would rise with order growth regardless of operations.

588,832

REVIEWS SCRAPED

Jun 2020 → Jul 2026

209,680

NEGATIVE BAND (1-2★)

35.6% of corpus

35,325

IN THE TWO DISCLOSED QUARTERS

15,885 + 19,440

52,664

SNIPPETS CLUSTERED

reviews split by clause

From corpus to analysis cohort

FILTERING, STEP BY STEP

POSITIVE 4-5★ · 355,083

NEGATIVE 1-2★ · 209,680

3★

Positive 60.3%

Negative 35.6%

Neutral 3★ 4.1% · 24,069

All reviews scraped

Jun 2020 – Jul 2026

588,832

Negative band only

1-2★, strict integer ratings

209,680

Two disclosed quarters

Jan–Jun 2026 only

35,325

THE TWO COHORTS COMPARED · SHOWN AT THEIR OWN SCALE

Q4 FY26

Jan–Mar 2026 · baseline

15,885

Q1 FY27

Apr–Jun 2026 · in question

19,440

SAMPLE FLOOR PER CATEGORY

n ≥ 30

Every one of the 25 categories cleared it in both quarters

QUARTERS MATCHED TO

Exact fiscal

Jan–Mar 2026 vs Apr–Jun 2026, as disclosed

Three things to know before the findings

01 Every number ahead is a share, not a count

Complaint counts rise automatically when orders rise 20.8%. Shares do not. Each figure is that category's percentage of the same quarter's negative reviews.

02 The two quarters are drawn the same way

Same source, same rating bands, same date logic. Whatever bias makes people leave reviews is present in both quarters, so it largely cancels in the comparison.

03 Only movement is interpreted, never level

Reviewers skew to extremes, so "7% of complaints mention refunds" says little about how common refund problems are. That 7% became 7.2% is the readable signal.

THE LIMIT WORTH STATING PLAINLY

× Reviews cannot price anything

Nothing here converts a complaint into rupees. The sizing on slide 11 is an explicitly bounded range with every assumption listed, not a point estimate dressed up as one.

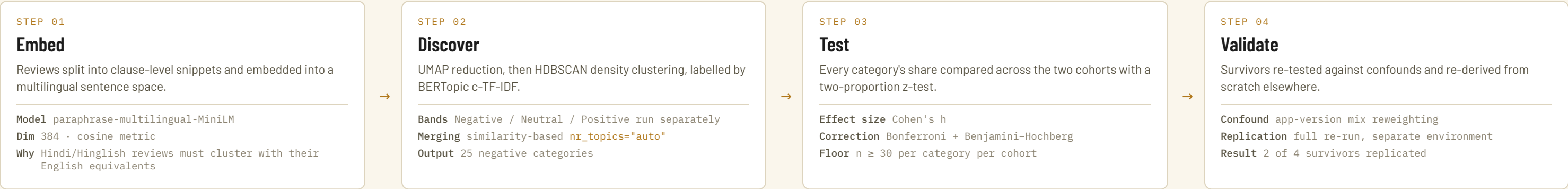
WHY THE CORPUS SUPPORTS THIS

Negative reviews per quarter roughly doubled over the series, from about 9,000 to 19,000, while the **negative band held steady near a third of all reviews**. The mix is measurable because the base is large and its composition is stable.

Data: Google Play Store reviews for the Blinkit Android app, scraped Jul 2026 (n = 588,832, Jun 2020 – Jul 2026). Bands are strict integer ratings (Negative 1-2★, Neutral 3★, Positive 4-5★), so no review appears in two bands. Reviews split into clause-level snippets before clustering. **Cohorts:** Q4 FY26 = Jan–Mar 2026 (n = 15,885). Q1 FY27 = Apr–Jun 2026 (n = 19,440).

Categories were discovered, not assumed, then made to survive **three independent attempts to kill them.**

No keyword list decides what a complaint is about. Reviews are embedded, clustered, and labelled from their own vocabulary. A category only reaches the findings if it clears significance **after** multiple-comparison correction, survives version-mix reweighting, and reappears in a re-run built in a different environment.



Choosing the clustering resolution

SEARCH

IN ONE LINE

This configuration won by a computed score, not because it happened to produce a nicer-looking result.

MIN CLUSTER	MIN SAMP	TOPICS	NOISE	SEPARATION
394 (0.75%)	100	27	22.3%	11.83
263 (0.50%)	100	30	20.6%	11.69
263 (0.50%)	75	32	20.7%	11.69
394 (0.75%)	75	29	20.8%	11.54

Selected by **highest inter/intra-cluster separation ratio** among configs with 6–25 topics and under 40% noise, not tuned after seeing which categories moved. A **fixed topic count was tried first and rejected**: it merged Cancellations and Refunds into one blob, while auto-merging kept them distinct.

The statistical machinery

PER CATEGORY

TEST STATISTIC

two-proportion z-test

Is category c's share of negative reviews different in Q1 FY27 than Q4 FY26?

EFFECT SIZE

$$h = 2 \cdot \arcsin \sqrt{p_1} - 2 \cdot \arcsin \sqrt{p_2}$$

Cohen's h, comparable across categories with very different base rates.

TIER 1

Bonferroni on the pre-registered branches, the conservative correction, applied where a hypothesis existed in advance.

TIER 2

Benjamini-Hochberg on categories discovery surfaced that no branch anticipated. Controls false-discovery rate without discarding real signal.

WHY

Applying one correction to both would either over-punish discovery or under-punish the pre-registered set.

The three gates a finding must pass

ALL OR NOTHING

1

Significant after correction

Adjusted p < 0.05 against the appropriate family. Four of 25 categories cleared this.

2

Robust to app-version mix

Q1 FY27 rates reweighted to Q4 FY26's version distribution. The effect must keep its sign and at least half its magnitude. Requires ≥50% version-mix overlap or the result is withheld.

3

Reproduced in an independent run

The entire pipeline re-executed in a separate environment. Clustering is not bit-identical across environments, so a finding that depends on one particular partition fails here.

Gate 3 is the one that did the most work: it demoted two of the four statistically significant findings. **Significance alone was not treated as evidence.**

Pipeline: sentence-transformers (paraphrase-multilingual-MiniLM-L12-v2) → UMAP (n_neighbors 15, 5 components, cosine) → HDBSCAN (EOM) → BERTopic c-TF-IDF labelling with similarity-based topic merging. **Statistics:** statsmodels two-proportion z-test, Cohen's h, Bonferroni and Benjamini-Hochberg correction, direct-standardisation reweighting on app-version groups. Resolution search over 5 min-cluster-size × 6 min-samples configurations, scored by inter/intra separation ratio.

25 complaint categories emerged from the corpus. The biggest are not the ones that moved.

Named by the model from its own top terms, then labelled in plain English. **Prevalence and change are different questions**: the largest category barely moved, and the one that doubled sits 20th by size.

All 25 discovered categories			NEGATIVE BAND · RANKED BY Q1 FY27 SHARE			
#	CATEGORY	MODEL-GENERATED TOP TERMS	Q4 FY26	Q1 FY27	← DECREASE · QOQ CHANGE · INCREASE →	QOQ
01	Delivery time & location	delivery, time, late, location	15.05%	14.75%		0.98×
02	General dissatisfaction	blinkit, experience blinkit, order, money	9.69%	11.17%		1.15× ●
03	App usability	app, worst app, worst, dont	10.03%	10.67%		1.06×
04	Hindi / Hinglish cluster	hai, bhi, nhi, se	9.73%	10.33%		1.06×
05	Customer support access	customer, support, customer support, care	7.00%	7.62%		1.09×
06	Missing / wrong items	missing, order, items, item	7.26%	7.25%		1.00×
07	Refunds & returns	refund, return, product, damaged	7.01%	7.20%		1.03×
08	Delivery & handling charges	charges, charge, delivery charges, handling	8.01%	6.72%		0.84× ●
09	Service quality (generic)	service, worst service, bad service, service service	5.80%	5.90%		1.02×
10	Product quality (generic)	quality, product, products, bad	3.97%	4.04%		1.02×
11	Pricing / too expensive	price, high, price high, prices	3.76%	3.63%		0.96×
12	Produce freshness (rotten)	vegetables, rotten, fruits, fresh	3.03%	3.62%		1.20× ●
13	Frustration language	frustrating, disappointed, unacceptable, happened	3.44%	3.48%		1.01×
14	Packaging & bags	bag, packaging, packing, dirty	3.02%	3.48%		1.15×
15	Payment & UPI failures	payment, upi, deducted, order	3.59%	3.42%		0.95×
16	Mixed good / bad	good, good good, bad, bad bad	2.95%	2.89%		0.98×
17	Order cancellation	cancel, cancel order, order, cancellation	2.96%	2.88%		0.97×
18	Bad experience (generic)	experience, experience bad, bad experience, worst experience	2.37%	2.69%		1.13×
19	Expired products	expired, expiry, date, expired products	2.11%	2.23%		1.06×
20	Cold-chain / melted items	ice, ice cream, cream, melted	0.90%	2.13%		2.37× ●
21	Coupons & offers	coupon, offers, discount, offer	2.12%	2.04%		0.96×
22	Cash on delivery	cash, cash delivery, delivery, delivery option	2.01%	1.95%		0.97×
23	Chatbot / AI support	chat, support, bot, ai	1.95%	1.94%		1.00×
24	Competitor comparison	zepto, better, instamart, flipkart	1.88%	1.62%		0.86×
25	Surge charges	surge, surge charge, charge, surge charges	1.48%	1.35%		0.91×

What the output tells us

- 01

Size and movement are unrelated
Delivery time is the largest category at 14.75% and moved 0.98×. Cold-chain is 20th at 2.13% and moved 2.37×. Ranking by volume would have found nothing.
- 02

Discovery split what a keyword list would merge
Produce freshness, expired products, packaging and cold-chain separated cleanly into four distinct failure modes that a single "damaged goods" tag would have averaged into noise.
- 03

Artefact categories are labelled, not hidden
"Hindi / Hinglish", "App usability" and "Frustration language" are linguistic or generic groupings, not operational failures. They are reported, but no cost claim is built on them.
- 04

Speed is a strength, not the problem
The same pipeline run on 4–5★ reviews returns one dominant praise theme: *fast delivery, quick, on time* (42,960 snippets). Customers are not telling us speed broke.
- 05

One visual conclusion
Almost every bar sits on the centre line. Only one category moved far enough to matter.

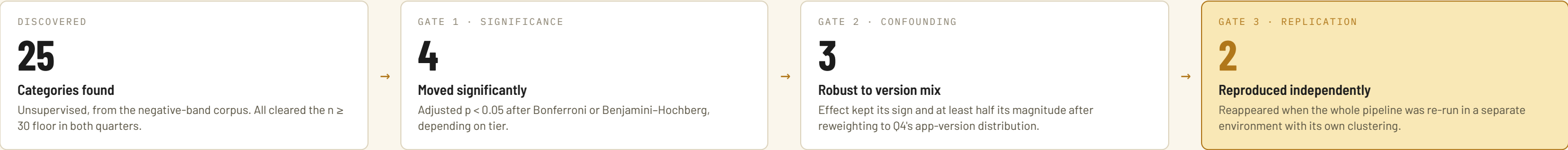
CATEGORY INTEGRITY

No category membership was hand-assigned. Labels are cosmetic renamings of model-generated term lists, applied **after** the statistics were computed, never used to move a review between categories.

● = significant after multiple-comparison correction (4 of 25). Change bar scaled to |QoQ – 1| with a maximum of 1.40. A review may match more than one category, so shares do not sum to 100%. **Source:** own BERTopic discovery over the Q4 FY26 + Q1 FY27 negative-review window (52,664 clause snippets from 35,325 reviews). Shares are the percentage of each quarter's negative reviews matched to a category by cosine similarity against category centroids, using per-category calibrated thresholds. Q4 FY26 n = 15,885. Q1 FY27 n = 19,440. Plain-English labels assigned by the author from each category's model-generated top terms.

Four categories moved significantly. Two survived every test.

Significance is where most analyses stop. Here it is only the first of three gates, and it eliminated the fewest candidates. Version-mix reweighting removed one finding, and independent replication removed another.



The four categories that moved, with every statistic behind them

Q4 FY26 → Q1 FY27 · NEGATIVE BAND

IN ONE LINE

Two of these four failed a later gate. The columns on the right show exactly which one, and why.

CATEGORY	Q4 FY26	Q1 FY27	QOQ	COHEN'S H	P (RAW)	P (ADJUSTED)	TIER	GATE 1	GATE 2	GATE 3	VERDICT
Cold-chain / melted items ice, ice cream, cream, melted, icecream, cold	0.90%	2.13%	2.37× ▲	0.103	2.1×10^{-20}	2.7×10^{-19}	Exploratory	✓	✓	✓	PRIORITY
General dissatisfaction blinkit, experience blinkit, order, money	9.69%	11.17%	1.15× ▲	0.048	7.1×10^{-6}	8.5×10^{-5}	Pre-registered	✓	✓	✓	CORROBORATING
Produce freshness (rotten) vegetables, rotten, fruits, fresh, quality	3.03%	3.62%	1.20× ▲	0.033	2.1×10^{-3}	2.5×10^{-2}	Pre-registered	✓	✓	✗	NOT CARRIED
Delivery & handling charges charges, charge, delivery charges, handling	8.01%	6.72%	0.84× ▼	0.049	3.5×10^{-6}	2.3×10^{-5}	Exploratory	✓	✗	✗	NOT CARRIED

GATE 2 DETAIL

Effect size in percentage points, before and after reweighting Q1 FY27 to Q4 FY26's app-version mix

Cold-chain

raw

+1.23pp

rewtd

+1.39pp

ROBUST

Sign held and magnitude grew. The rise is not an artefact of which app version users were on.

General dissatisfaction

raw

+1.47pp

rewtd

+1.96pp

ROBUST

Sign held and magnitude grew. Survives, but names no single fixable mechanism.

Produce freshness

raw

+0.59pp

rewtd

+0.87pp

ROBUST

Passed this gate, then failed replication in the independent re-run (adjusted p = 0.065).

Delivery & handling charges

raw

-1.29pp

rewtd

-0.32pp

NOT ROBUST

Three-quarters of the decline disappeared on reweighting. Mostly app-version mix, not sentiment.

Statistics: two-proportion z-test on category share of negative reviews, Q4 FY26 (n = 15,885) vs Q1 FY27 (n = 19,440). Cohen's h reported as absolute magnitude with direction shown by arrow. Adjusted p-values use Bonferroni within the pre-registered tier and Benjamini-Hochberg within the exploratory tier. **Gate 2:** direct-standardisation reweighting on app-version groups, requiring ≥50% version-mix coverage. **Gate 3:** full pipeline re-executed independently. See slide 15 for the side-by-side comparison.

Cold-chain failure: complaints more than doubled, and customers describe a temperature problem, not a service problem.

This is the only category that is large in effect, significant after correction, unconfounded by app-version mix, and reproducible in an independent run. The verbatims name the mechanism directly: packaging, temperature control, and dwell time. That is what makes it actionable rather than merely correlated.

The movement

QOQ

2.37×

RISE IN SHARE OF NEGATIVE REVIEWS

Q4 FY26

0.90%

Q1 FY27

2.13%

Complaints, Q4 FY26

143

Complaints, Q1 FY27

415

Adjusted p-value

2.7 × 10⁻¹⁹

Cohen's h

0.103

Rank by size (of 25)

20th

Rank by movement

1st

TRIANGULATION ACROSS RATING BANDS

Negative 1-2★

2.13%

Neutral 3★

1.09%

Positive 4-5★

0.26%

The theme concentrates sharply in the negative band, consistent with a genuine failure rather than a topic customers simply mention more often.

What customers actually wrote

VERBATIM • Q1 FY27

"It seems like there was no proper temperature control during delivery."

"The upper cover of the ice cream was also missing, and it was in a torn and melted condition."

"Frozen items like icecream are not well packed."

"at last i received only the ice cream juice"

"After sometime we found the the tub was broken from bottom."

"one packet of milk was damaged, I returned and he promised to bring new one."

"Fast delivery means nothing if basic quality standards are ignored."

"received 3 ice-cream instead of 4."

↳ audited: missing item, not a cold-chain failure

"And the customer care is just a joke."

↳ audited: support complaint, not a cold-chain failure

All nine drawn at random from the 601 Q1 FY27 reviews in this category, **not hand-picked**. Representative-document sampling was avoided because it favours centroid-proximal text and hides minority sub-themes. The last two are shown deliberately: **they are the contamination**, quantified in the caveat opposite rather than quietly removed.

Why this lands in direct cost per order

MECHANISM

1

Item degrades between freezer and doorstep

Blinkit's 10–15 minute model has no refrigerated transit. Cold retention depends on dark-store storage, packaging and dwell time.

2

The unit is written off

A melted or spoiled item cannot be resold or restocked.
→ INVENTORY WASTAGE

3

A replacement is dispatched

A second rider trip for the same revenue, so the delivery cost is incurred twice on one order.
→ DELIVERY COST, DOUBLED

4

Or the order is refunded

Full or partial credit issued, plus a support contact to process it.
→ CREDIT + SUPPORT COST

Σ

Direct cost per order rises

One failure mode touches three separate cost lines simultaneously, which is why a category this small by volume can move a blended per-order number.

! Most of this is seasonal

Tested on the next slide: the Apr–Jun spike recurs every year. Roughly 80% of the rise is the normal summer pattern and 20% is excess. The mechanism below is unchanged, but the scale attributable to operations is smaller.

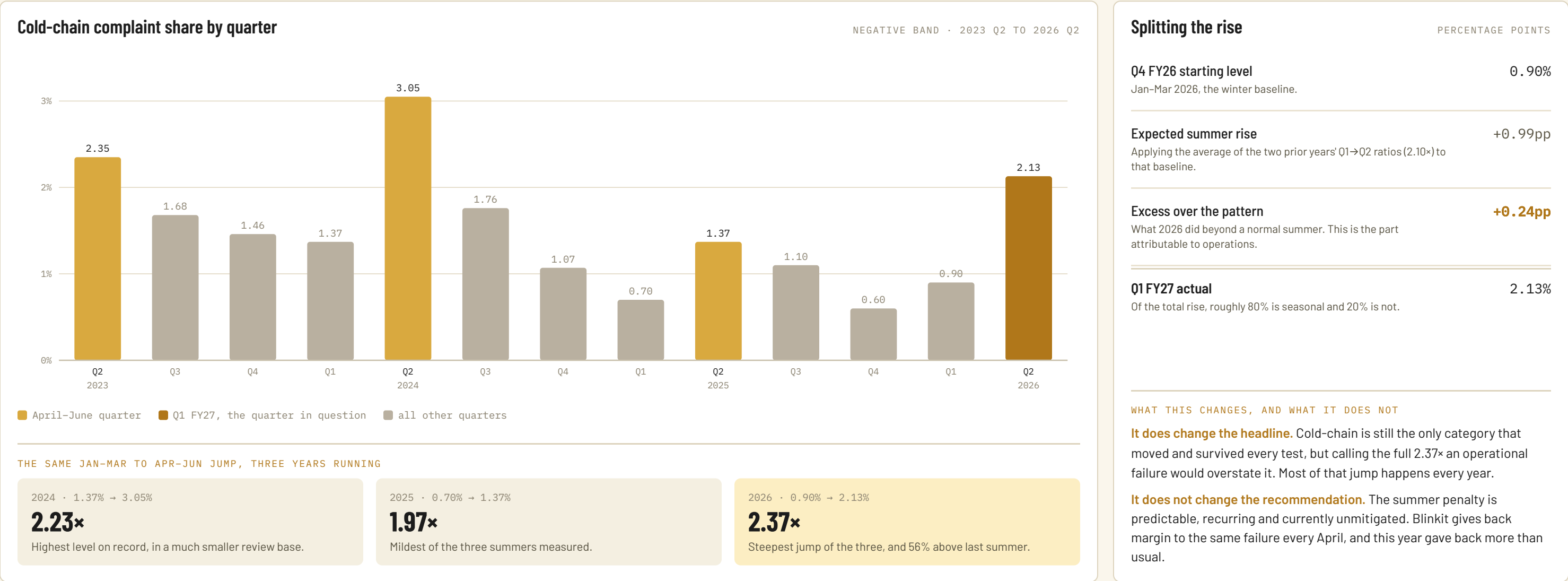
! The cluster is not perfectly clean

Random sampling found unrelated complaints inside it: missing items and support failures. The signal is real but its magnitude is an upper bound on pure cold-chain complaints.

Category: discovered cluster with model-generated terms *ice*, *ice cream*, *cream*, *melted*, *icecream*, *cold*, *chilled*. 143 of 15,885 Q4 FY26 negative reviews and 415 of 19,440 Q1 FY27 negative reviews matched. **Quotes:** randomly sampled from the 601 Q1 FY27 reviews assigned to the category via cosine similarity, filtered only for length. Band shares from the same category centroid applied to the Neutral (3★) and Positive (4–5★) discovery runs.

Cold-chain complaints spike every April. Most of the rise is seasonal, and that is the more useful finding.

Extending the same measurement back three years shows the Apr–Jun peak is not new: it has happened in every year on record. Roughly **four-fifths of this quarter's rise is the normal summer pattern**, and about one-fifth is genuine deterioration on top of it. A cost that arrives on schedule every year is easier to plan for, not harder.



The opportunity is bounded, not guessed, and it sets a budget of roughly ₹5.65 per frozen order for any fix.

Blinkit does not disclose a wastage cost line, so cold-chain's rupee share of the ₹74.6cr rate effect cannot be computed directly. What can be done is bound it, state every assumption, and convert the result into the only number an operator needs: **how much the fix is allowed to cost**.

From cost rise to attributable opportunity

PER QUARTER

ESTABLISHED ON SLIDE 03 • ARITHMETIC, NOT ESTIMATE

₹74.6 cr

The rate effect: the portion of the direct-cost increase caused by per-order efficiency deteriorating rather than by serving 20.8% more orders.

10%	₹7.5 cr	₹30cr annualised
25%	₹18.7 cr	₹75cr annualised
50%	₹37.3 cr	₹149cr annualised

₹74.6CR IN CONTEXT

As a share of Q1 FY27 total direct cost (₹3,806cr)		2.0%
Against Blinkit's Q1 FY27 adjusted EBITDA (₹102cr)		73%
The 25% anchor (₹18.7cr) against that same EBITDA		18%

Why 25% is the working anchor: cold-chain is the largest validated movement in the entire negative corpus, and it touches three cost lines at once: wastage, redelivery and refund. Claiming under 10% would contradict the effect size. Claiming over 50% would require cold-chain to dominate a cost base that also carries rider pay, packaging and support.

Converting the opportunity into a spend limit

BREAK-EVEN

MAXIMUM JUSTIFIABLE SPEND

₹5.65

per order containing a frozen or chilled item, assuming those are 10% of orders and the 25% attribution holds. Any intervention cheaper than this pays for itself within the quarter.

IF FROZEN/CHILLED ORDERS ARE...	BUDGET PER SUCH ORDER
5% of all orders	₹11.30
10% of all orders	₹5.65
15% of all orders	₹3.77
20% of all orders	₹2.82
Spread across all 331M orders	₹0.56

The threshold is deliberately conservative. It credits the fix only with avoided wastage, not with the redelivery trips, refunds and support contacts the same failure also generates. A fix that clears this bar clears it with margin.

This is a threshold, not a cost estimate. It converts the opportunity into a per-unit allowance, tested directly on slide 13 against real bulk gel-pack pricing (₹3–9/piece, IndiaMART): the lower half of that range clears this line, the upper half does not.

Assumption register

WHAT EACH NUMBER RESTS ON

₹115 direct cost per order

Disclosed by Eternal for Q1 FY27.

DISCLOSED
no assumption

+2.0% QoQ rate rise

Disclosed. Q4 FY26 level implied by reversing it.

DISCLOSED
no assumption

331M orders in Q1 FY27

From Q1 FY27 investor-deck coverage, cross-checked against 20.8% QoQ and 87% YoY growth.

REPORTED
cross-checked

₹74.6cr rate effect

Arithmetic on the three lines above. No judgement involved.

DERIVED
deterministic

25% cold-chain attribution

Judgement, bounded by a 10–50% range. The single largest source of uncertainty in this case.

ASSUMED
±2× either way

10% of orders carry frozen items

Illustrative only, used to express the budget per affected order. Full sensitivity shown opposite.

ASSUMED
illustrative

Seasonality contribution

Measured on slide 10. About 80% of the cold-chain rise is the recurring summer pattern, so the opportunity is a recurring annual saving rather than a one-off recovery.

TESTED
see slide 10

■ disclosed / derived ■ assumed ■ untested

Rate effect: (₹115 – ₹112.75) × 331M orders = ₹74.6cr, established on slide 03. **Attribution:** the 10 / 25 / 50% range is the author's judgement, not a measured split. Eternal does not disclose a wastage or cold-chain cost line. **Break-even:** attributable opportunity ÷ (331M × assumed frozen-order share). Annualised figures assume the quarterly rate persists and are shown for scale only, not as a forecast.

Late delivery, cancellations, refunds and wrong items are flat. The obvious suspects are not the story.

These complaints are frequent, loud and entirely real. They simply did not **change** between the two quarters, and a cost line only moves when something changes. Reporting them as null is a finding in itself. It removes four expensive remediation programmes from consideration.

Every pre-registered branch, with its verdict

Q4 FY26 → Q1 FY27

IN ONE LINE

Eleven branches, eleven non-events. This table exists to prove that, not just assert it.

GENUINELY FLAT	movement indistinguishable from zero				
CATEGORY	Q4 FY26	Q1 FY27	QOQ	P (RAW)	P (ADJ)
Chatbot / AI support	1.95%	1.94%	1.00×	0.996	1.000
Missing / wrong items	7.26%	7.25%	1.00×	0.952	1.000
Delivery time & location	15.05%	14.75%	0.98×	0.442	1.000
Order cancellation	2.96%	2.88%	0.97×	0.643	1.000
Refunds & returns	7.01%	7.20%	1.03×	0.478	1.000
Expired products	2.11%	2.23%	1.06×	0.428	1.000
Product quality (generic)	3.97%	4.04%	1.02×	0.754	1.000
MOVED, BUT NOT ENOUGH TO SURVIVE CORRECTION reported honestly, not claimed					
Packaging & bags	3.02%	3.48%	1.15×	0.017	0.072
Customer support access	7.00%	7.62%	1.09×	0.026	0.306
Bad experience (generic)	2.37%	2.69%	1.13×	0.064	0.119
App usability	10.03%	10.67%	1.06×	0.050	0.119
WHAT THIS REMOVES FROM THE Q2 AGENDA					
× Rider-speed programmes, delivery complaints fell slightly			× Support headcount expansion, contact complaints not significant		
× Cancellation-flow redesign, 0.97× and moving the wrong way to matter			× Refund-speed initiatives, 1.03× and inside the noise band		

These complaints are real

VERBATIM

LATE DELIVERY • 0.98×

"delivery is always late, it shows 13min delivery time, but after 13min it still showing 13min delivery time."

CUSTOMER SUPPORT • 1.09×, NOT SIGNIFICANT AFTER CORRECTION

"No customer support."

CANCELLATIONS • 0.97×

"CAN'T CANCEL MY ORDER"

REFUNDS • 1.03×

"once done, they never refund."

MISSING ITEMS • 1.00×

"items were missing in the order."

Nothing here says these experiences are acceptable. It says they were **equally bad in both quarters**, so they cannot explain a cost line that moved in only one of them.

Was the study powerful enough to see a real effect?

POWER

IN ONE LINE

The test could catch a move as small as +30%. Cold-chain moved +137%, so this was never a close call.

MINIMUM DETECTABLE EFFECT • 80% POWER

Cohen's h = 0.040

After Bonferroni correction across the pre-registered branches. Anything larger than this would have been caught.

AT A BASELINE OF...	DETECTABLE FROM	RELATIVE
2% (e.g. cold-chain)	2.59%	+30%
7% (e.g. refunds)	8.05%	+15%
15% (e.g. delivery)	16.44%	+10%

01 Cold-chain cleared the bar by a wide margin

It moved +137% against a +30% detection threshold, roughly four and a half times the minimum detectable effect.

02 The flat categories are genuinely flat

Delivery, cancellations, refunds and missing items all moved under 6% against a 10–15% detection floor. These are not underpowered tests. The effects are simply absent.

03 Two categories sit in the grey zone

Packaging (+15%) and customer support (+9%) are near or below the correction-adjusted threshold. They are reported as **not detected**, not as **no effect**. Packaging is worth watching, since it shares a mechanism with cold-chain.

Power calculation: two-proportion test, n = 15,885 and 19,440, 80% power, two-sided α = 0.05 Bonferroni-corrected across 12 pre-registered branch tests (α = 0.0042). Minimum detectable Cohen's h = 0.040. The uncorrected figure is 0.030. **p-values:** raw and adjusted values as computed in the pipeline. Adjusted values of 1.000 indicate Bonferroni capping. All categories cleared the $n \geq 30$ floor in both quarters.

Fix the cold chain, not "quality". Three interventions, each interrupting a specific link in the mechanism.

Every action below targets the chain established on slide 09: item degrades → written off → redelivered or refunded. Each is costed against the ₹5.65 per frozen order break-even from slide 11, and each has a single metric that would show it working within one quarter.

1DO FIRST · HIGHEST RATIO

Insulated liners and gel packs for frozen & dairy SKUs

Mandated at dark-store pack-out for any basket containing a frozen or chilled item, not offered as an option to the picker.

⌘ INTERRUPTS LINK 1 · degradation between freezer and doorstep

WHY THIS FIRST

Customers named packaging directly: **"Frozen items like icecream are not well packed."** It requires no route change, no headcount and no software release.

COST AGAINST THE BREAK-EVEN

Break-even		₹5.65
Bulk price		₹3-9

Gel-pack inserts, IndiaMART bulk listings (Aug 2026). **The lower half of this range clears the break-even, the upper half does not.** The case holds if procurement can land below ~₹5.65 at 331M-order scale, well above the 1,000-5,000-piece minimums these listings reflect.

PRIMARY METRIC

Cold-chain complaint share, weekly

Target: back under 1.2% within one quarter, from 2.13%.

2DO NEXT · DIAGNOSTIC

Dark-store freezer uptime and capacity audit

Instrument freezer temperature and door-open time across the 2,443 stores, ranked by deviation and by frozen-SKU throughput per unit of freezer volume.

⌘ INTERRUPTS LINK 1 · upstream, before the rider is involved

WHY IT MATTERS HERE

A 10-15 minute model has **no refrigerated transit**. Cold retention is inherited entirely from the store. 200 net new stores opened this quarter, a plausible source of variance in exactly this window.

WHAT IT ANSWERS

Whether this is a **packaging problem** or a **refrigeration-capacity problem**. The two have very different capex profiles, and slide 09's evidence cannot separate them.

PRIMARY METRIC

% of stores outside target freezer band

Correlate store-level deviation against store-level cold-chain complaint rate.

3THEN · SYSTEMIC

Separate dwell-time SLA for frozen items in mixed baskets

Frozen picks sequenced last and dispatched on their own clock, rather than waiting in ambient conditions for the rest of a multi-item order.

⌘ INTERRUPTS LINK 1 · dwell time inside the store

Highest effort of the three: it touches picking logic and the dispatch algorithm. Sequenced last because interventions 1 and 2 will reveal how much dwell time actually contributes.

GUARDRAIL

Must not degrade overall delivery time, the one thing customers currently **praise** (42,960 positive mentions of fast delivery). Speed is an asset to protect, not trade away.

PRIMARY METRIC

Frozen-item dwell time, p90

Paired guardrail: overall on-time delivery must not regress.

THE ONE-LINE CASE

A **₹3-9 gel-pack insert** (IndiaMART bulk pricing) against a **₹5.65 allowance**, targeting the only complaint category that doubled, survived every statistical gate, and reproduced independently. The bottom half of that range clears the allowance with margin, and the top half does not. The case rests on negotiating into the lower band, which is realistic at 331M orders/quarter.

EXPLICITLY NOT RECOMMENDED

Nothing in the evidence supports funding these this quarter:

Rider-speed programmes

Support headcount

Cancellation-flow redesign

Refund-speed initiatives

TIMING MATTERS

The spike is seasonal and arrives every April (slide 10). Liners need to be in stores **before the season starts**, not in response to it. Deploying in July concedes another year of the same cost.

Break-even: ₹5.65 per frozen order, from slide 11: attributable opportunity ÷ (331M orders × 10% assumed frozen share). **Gel-pack pricing:** IndiaMART bulk listings, Aug 2026: Blucia Overseas ₹3/piece, Shakya World Trade ₹6/piece, Valor Industries ₹9/piece, at listed minimum order quantities of 1,000-5,000 pieces. No Blinkit-specific quote was obtained; reusable insulated bags (~₹944/unit, TradeIndia) are a separate amortized capex item and are not part of this per-order comparison. **Store count:** 2,443 dark stores at Q1 FY27, +200 net new in the quarter, per investor-deck coverage. **Positive-band mentions:** 42,960 snippets in the dominant fast-delivery praise theme from the same pipeline run on 4-5★ reviews.

Pilot it as a controlled experiment, held to the same standard of evidence used to find the problem.

The diagnosis rests on a between-cohort comparison with corrections and replication. The fix should be held to the same bar: **treated and control dark stores, a pre-declared success threshold, and a stop rule written before the pilot starts.**

Sequencing

IMPACT VS EFFORT

STOP RULE • DECLARED BEFORE THE PILOT

×

 Treated-vs-control gap in cold-chain complaint share is not significant at 6 weeks → **do not scale**, escalate to intervention 2.

×

 Overall on-time delivery regresses in treated stores → **halt**. Speed is the asset customers actually praise.

×

 Procurement quote exceeds ₹5.65 per frozen order → **re-open the business case**, do not proceed on faith.

Twelve weeks, four phases, one decision gate each

FROM QUOTE TO NATIONAL ROLLOUT

WEEKS 0–2

Quote & design

01

 Obtain liner and gel-pack quotes at national volume

02

 Select ~40 dark stores, split 20 treated / 20 control, matched on frozen-SKU mix and city

03

 Baseline each store's cold-chain complaint share over the prior 8 weeks

GATE

Unit cost lands under ₹5.65

WEEKS 2–8

Run the pilot

01

 Mandatory liners on every frozen basket in treated stores only

02

 Controls change nothing: no communication, no incentives

03

 Weekly complaint-share read on both arms, same pipeline as this analysis

04

 Track on-time delivery as a guardrail in both arms

GATE

Treated arm's complaint share falls significantly vs control

WEEKS 8–10

Read & attribute

01

 Two-proportion test on treated vs control, same correction discipline

02

 Convert the observed drop into a wastage-cost estimate using internal data, closing the gap this case could only bound

03

 Compare the treated arm against the same stores in the prior summer, not just against control

GATE

Payback confirmed against real cost data

WEEKS 10+

Scale or stop

01

 If confirmed: roll to all 2,443 stores, frozen baskets first

02

 Fold cold-chain complaint share into the weekly ops dashboard as a standing metric

03

 If not confirmed: proceed to the freezer audit before any further spend

GATE

Complaint share sustained under 1.2%

WHY A CONTROL ARM IS NON-NEGOTIABLE HERE

THE CONFOUND

Seasonality

Complaints may fall from July simply because the weather cools.

WITHOUT CONTROLS

Unattributable

A post-only drop cannot separate the liner from the season.

WITH CONTROLS

Clean read

Both arms experience the same weather, and only the liner differs.

COST OF THE RIGOUR

Two weeks

The delay that makes the result defensible to finance.

Pilot sizing: 20 treated and 20 control stores is indicative. The final count should be powered against the observed store-level variance in cold-chain complaint share, using the same minimum-detectable-effect calculation shown on slide 12. **Target:** cold-chain complaint share returning below 1.2%, roughly the midpoint between the Q4 FY26 level (0.90%) and the Q1 FY27 level (2.13%). **Guardrail metric:** on-time delivery rate, to protect the strength identified in the positive-band analysis.

The whole pipeline was run twice, independently. Half the significant findings did not survive it.

Clustering is not deterministic across environments. A finding that depends on one particular partition of the corpus is not a finding. Running the analysis a second time from scratch was the single most informative check performed, and it is the reason this deck carries one recommendation rather than four.

Same data, two independent runs			
LOCAL VS CLOUD ENVIRONMENT			
FINDING	RUN A · LOCAL	RUN B · CLOUD	VERDICT
Cold-chain / melted items ice, ice cream, melted, cold	2.68× SIG · ROBUST	2.37× SIG · ROBUST	REPLICATES
General dissatisfaction blinkit, experience, money	1.14× SIG · ROBUST	1.15× SIG · ROBUST	REPLICATES
Produce freshness (rotten) vegetables, rotten, fruits	1.18× NOT SIGNIFICANT	1.20× SIG · ROBUST	CONTRADICTS
Delivery & handling charges charges, handling, extra	0.86× SIG · NOT ROBUST	0.84× SIG · NOT ROBUST	CONSISTENTLY WEAK
FINDINGS THAT APPEARED IN ONLY ONE OF THE TWO RUNS			
Packaging & milk damage milk, bag, packaging, dirty	1.21× · SIG	not significant	discarded
App usability app, worst app, dont, use	1.08× · SIG	not significant	discarded
Produce freshness vegetables, rotten, fruits	not significant	1.20× · SIG	discarded
WHAT AGREED AND WHAT DID NOT			
REVIEW COUNTS	CATEGORIES FOUND	TOP FINDING	SURVIVED BOTH
Identical 15,885 and 19,440 in both runs	22 vs 25 clustering differs by environment	Same cold-chain first by movement in both	2 of 4 the two carried into this deck

Why two runs disagree at all

- 01

UMAP is not reproducible across machines
Even with a fixed random seed, thread count and library versions change the embedding layout. Different layouts produce different cluster boundaries.
- 02

Cluster count changes the correction
22 versus 25 categories changes the Bonferroni family size, which moves borderline p-values across the 0.05 line. Produce freshness sat exactly there.
- 03

Membership boundaries shift
A snippet near two centroids can land in either. For a large effect this is irrelevant. For a small one it is decisive.
- 04

The data layer was perfectly stable
Scraping, banding and quarter assignment produced byte-identical cohorts. Every discrepancy originates in clustering, nowhere else.

This is the argument for treating replication as a gate rather than a nicety: **a single run would have produced a four-finding deck, and two of those findings would have been wrong.**

Limitations register

- Seasonality now measured**
The Apr–Jun spike recurs in every year on record. About 80% of this quarter's rise is that pattern, so the operational share is roughly one-fifth of the headline number.

RESOLVED
slide 10
- Attribution share is judgement**
The 25% anchor converting complaints into ₹18.7cr is bounded but not measured. Internal wastage data would replace it.

HIGH
bounded
- Cluster contains contamination**
Random sampling found unrelated complaints inside the cold-chain category, so 2.13% is an upper bound on pure cold-chain mentions.

MEDIUM
quantified
- Reviewers are self-selected**
Mitigated by comparing composition between two equally-biased quarters rather than reading levels, but not eliminated.

MEDIUM
mitigated
- Android only**
iOS reviews were not scraped. No reason to expect a different cold-chain pattern, but it is untested.

LOW
untested

WHAT WOULD CHANGE THE CONCLUSION

- A fourth summer comes in below the 2.10× baseline → the 2026 excess was noise
- Internal wastage cost is flat QoQ → complaints rose without cost following
- A third independent run loses cold-chain → treat as unstable

Run A: local execution, 22 negative categories discovered. **Run B:** independent cloud execution of the same notebook on the same input data, 25 negative categories discovered. Both runs used identical cohort definitions and produced identical review counts (15,885 and 19,440). Category identifiers are not comparable across runs, so findings were matched by their model-generated top terms. Full per-category outputs for both runs are retained.

One quarter, one cost line, one defensible answer.

The value of this case is not that it found something. It is that **it discarded almost everything it found**. Twenty-five categories became four, four became two, and only one names a mechanism an operator can act on this quarter.

The argument, end to end

● THE TRIGGER

₹115 direct cost per order, +2.0% QoQ

Costing 10bps of contribution margin in Q1 FY27.

● THE ISOLATION

Only ₹74.6cr of ₹718cr is efficiency loss

Nine-tenths of the cost increase is growth. **The rate effect is the diagnosable part.**

● THE EVIDENCE

Cold-chain complaints rose 2.37×

0.90% → 2.13% of negative reviews. Significant, unconfounded, and **reproduced in a second independent run.**

● THE SEASONALITY CHECK

Four-fifths of that rise happens every summer

Measured back three years. The residual is **real but smaller than the headline**, and the pattern itself is the actionable part.

● THE ELIMINATION

Delivery, support, refunds, cancellations are flat

All within the noise band, against a detection floor that would have caught a 10–15% move.

● THE ACTION

Insulated liners, piloted with controls

A **₹3–9** gel-pack insert, priced against a **₹5.65** break-even, with a stop rule written before it starts.

What happens next

TWELVE WEEKS

W0

Get a liner quote at national volume

The entire business case turns on whether the unit cost lands under ₹5.65 per frozen order. Procurement can answer this in a fortnight.

W2

Run 20 treated stores against 20 controls

Matched on frozen-SKU mix and city. Controls change nothing, so the summer affects both arms equally and the liner is the only difference.

W8

Read the gap, then price it internally

Convert the observed drop into wastage cost using internal data. That closes the one gap this analysis could only bound from outside.

W10

Scale before the next season, not after

The spike arrives every April. Rolling out in response to it concedes another year of the same cost.

ONGOING

Keep the measure running

Cold-chain complaint share is cheap to track weekly and moves months before a quarterly cost line does.

How this was produced

REPRODUCIBILITY

Data

588,832 reviews · Jul 2026

Discovery

BERTopic, unsupervised

Testing

two-proportion z-test

Correction

Bonferroni + BH

Confound check

version-mix reweighting

Replication

full independent re-run

Seasonality

13-quarter back-test

Power

min detectable h = 0.040

STACK

Python

sentence-transformers

UMAP

HDBSCAN

BERTopic

statsmodels

pandas

PyArrow

The finding that mattered was 20th by volume and 1st by movement.

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25 → 1

CATEGORIES TO RECOMMENDATION